DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Linear Regression: Estimation

Learning Objectives

- Set up Linear Regression Models
- Estimate Linear Regression Models using Least Square
  - Simple Linear Regression
  - Multiple Linear Regression

Linear Regression Model

- Consider a regular linear regression model

$$y_i = \beta_0 + \beta_1 \cdot x_{i1} + \cdots + \beta_k \cdot x_{ik} + u_i$$

  - y_i: dependent (response/outcome) variable
  - x_i: independent (explanatory/predicator) variables
  - u_i: error term (unobservable idiosyncratic error)

$$E(u_i|X) = 0, \qquad \mathrm{Var}(u_i|X) = \sigma^2$$

  - β: coefficients (model parameters to be estimated)

Linear Regression Model (Cont'd)

market	sales	TV	radio	newspaper
1	22.1	230.1	37.8	69.2
2	10.4	44.5	39.3	45.1
3	9.3	17.2	45.9	69.3
4	18.5	151.5	41.3	58.5
5	12.9	180.8	10.8	58.4
6	7.2	8.7	48.9	75
7	11.8	57.5	32.8	23.5
8	13.2	120.2	19.6	11.6
9	4.8	8.6	2.1	1
10	10.6	199.8	2.6	21.2

$$sales_i = \beta_0 + \beta_1 \cdot TV_i + \beta_2 \cdot radio_i + \beta_3 \cdot newspaper_i + u_i$$

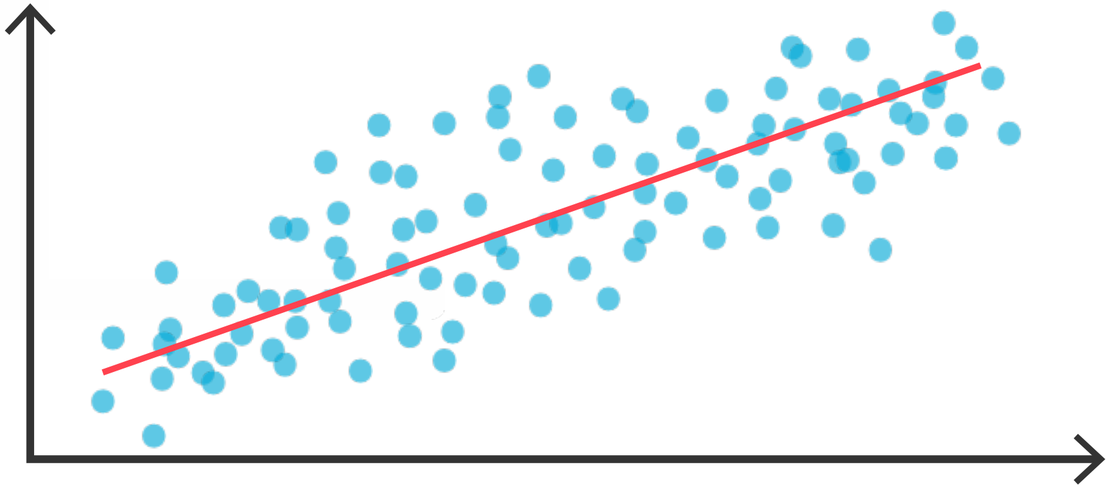
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_k x_{ik}$$

$$\widehat{sales}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot TV_i + \hat{\beta}_2 \cdot radio_i + \hat{\beta}_3 \cdot newspaper_i$$

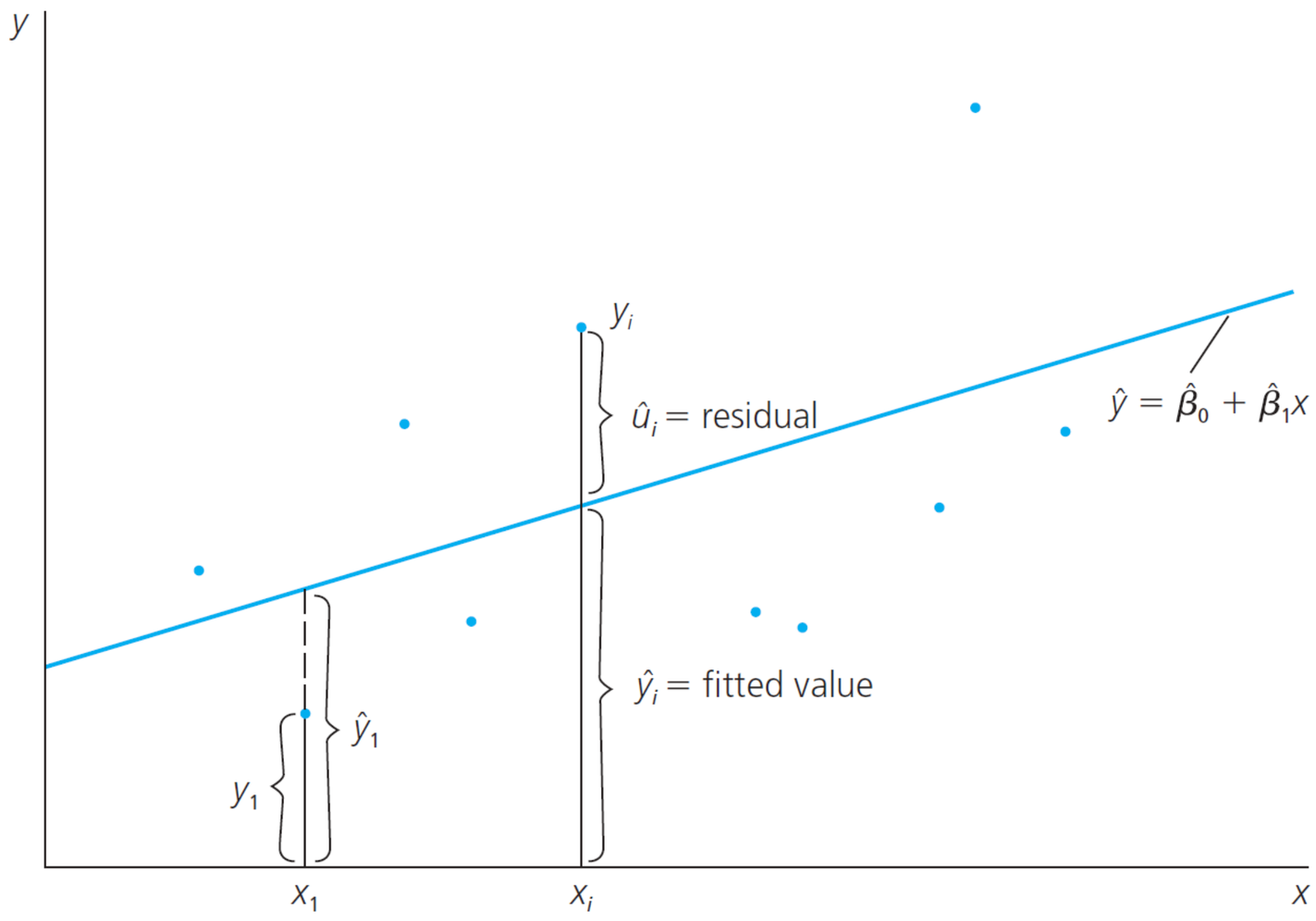
Estimation: Least Square (1/4)

- Estimation: Ordinary Least Square (OLS)

$$\min_{\beta_0,\ldots,\beta_k} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 \cdot x_{i1} - \cdots - \beta_k \cdot x_{ik})^2$$



Estimation: Least Square (2/4)



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Estimation: Least Square (3/4)

- <u>Simple Linear Regression</u>

$$y_i = \beta_0 + \beta_1 \cdot x_i + u_i$$

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 \cdot x_i)^2$$

$$\begin{cases} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \\ \sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \end{cases}$$

$$\Rightarrow \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\Rightarrow \quad \sum_{i=1}^{n} x_i [y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i] = 0$$

$$\sum_{i=1}^{n} x_i (y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^{n} x_i (x_i - \bar{x})$$

$$\begin{cases} \sum_{i=1}^{n} x_i (y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \\ \sum_{i=1}^{n} x_i (x_i - \bar{x}) = \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) \end{cases}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{Cov(x, y)}{Var(x)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Estimation: Least Square (4/4)

- <u>Multiple Linear Regression</u>

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

$$\min_{\beta_0,\ldots,\beta_k} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_k x_{ik})^2$$

$$\begin{cases} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0 \\ \sum_{i=1}^{n} x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0 \\ \cdots \\ \sum_{i=1}^{n} x_{ik}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik}) = 0 \end{cases} \Rightarrow \quad \mathbf{X}'(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}) = \mathbf{0}$$

$$\mathbf{X}'\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$$

$$\Downarrow$$

$$\boxed{\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}}$$

Summary

- Linear Regression Model Setup
- Least Square Estimation
  - Simple Linear Regression
  - Multiple Linear Regression

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Linear Regression: Interpretation and Evaluation

Learning Objectives

- Interpret Coefficients
  - Marginal Effects
  - Partialling Out Interpretation
- Measure the goodness-of-fit
  - R-Squared
  - Adjusted R-Squared

Interpreting Coefficients

- <u>How to Interpret Regression Coefficients:</u>

$$y_i + \Delta y = \beta_0 + \beta_1(x_{i1} + \Delta x_1) + \cdots + \beta_k x_{ik} + u_i$$

$$\Delta y = \beta_1 \Delta x_1$$

- Partial Effects (Marginal Effects)
  - β_1 measures the change in y due to one-unit change in x_1, <u>*holding all other independent variables unchanged*</u>.
  - Ceteris paribus interpretation
  - We have *controlled for* $x_2, \ldots, x_k$

Interpreting Coefficients (Cont'd)

<u>"Partialling Out" Interpretation</u>

$$\hat{\beta}_j = \frac{\sum_{i=1}^{n} \hat{r}_{ij} y_i}{\sum_{i=1}^{n} \hat{r}_{ij}^2}$$

- $\hat{r}_{ij}$ are the residuals from a regression of x_j on all the *other* x variables
- $\hat{\beta}_j$ can be obtained from a simple regression of y on $\hat{r}_j$

- $\hat{r}_{ij}$ is the residual of x_j after the effects of all the other x variables have been ***partialled out*** (or ***netted out***)
- β_j measures the effect of x_j on y <u>after all the other independent variables have been partialled out</u>.

Goodness of Fit (1/3)

- Total Sum of Squares (TSS)

$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- Explained Sum of Squares (ESS)

$$ESS = \sum_{i=1}^{n} \left(\hat{y}_i - \bar{\hat{y}}\right)^2$$

- Residual Sum of Squares (RSS)

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \hat{u}_i^2$$

- It can be shown that

$$TSS = ESS + RSS$$

Goodness of Fit (2/3)

<u>R-Squared</u>

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$$

- R^2 measures the proportion of the sample variation in y that can be explained by the linear regression.
- R^2 is between zero and one; *<u>the larger, the better model fit</u>*
- R^2 can also be shown as **the square of the correlation coefficient between y and $\hat{y}$**

$$R^2 = \frac{Cov(y, \hat{y})^2}{Var(y) \cdot Var(\hat{y})}$$

- Adding more independent variables always increases R^2 (because it would reduce RSS)

Goodness of Fit (3/3)

<u>Adjusted R-Squared</u>

$$\bar{R}^2 = 1 - \frac{RSS/(n-k-1)}{TSS/(n-1)}$$

$$= 1 - (1 - R^2)\frac{n-1}{n-k-1}$$

- $\bar{R}^2$ penalizes the number of parameters in the model
- Can be used to compare different models
- $\bar{R}^2$ can be negative!
- **Note**: We cannot use either R^2 or $\bar{R}^2$ to compare models with different functional forms of the *dependent* variable

Summary

- Coefficient Interpretation
  - Marginal Effects
  - Partialling Out Interpretation
- Model goodness-of-fit
  - R-Squared
  - Adjusted R-Squared

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Linear Regression: *t*-Test

Learning Objectives

- Explain the variance of the estimators
- Estimate the standard error of the estimators
- Perform *t*-Test
  - Compute the *t*-statistic
  - Find the *t*-threshold
  - Compute the *p*-value
- Detect multi-collinearity
  - VIF

Variance of the Estimators

<u>Variance of $\hat{\beta}$:</u>

- $\hat{\beta}$ is <u>a function of the data</u>
- Given the data, $\hat{\beta}$ is fixed
- However, the data sample we draw is random, so $\hat{\beta}$ is a random variable

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{\sum_{i=1}^{n} \hat{r}_{ij}^2}$$

$$= \frac{\sigma^2}{TSS_j(1 - R_j^2)}$$

Variance of the Estimators (Cont'd)

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{\sum_{i=1}^{n} \hat{r}_{ij}^2}$$

$$= \frac{\sigma^2}{TSS_j(1 - R_j^2)}$$

- σ^2 is the variance of the error term
- $\hat{r}_{ij}$ are the residuals from regressing x_j on all the other independent variables (including the intercept)
- R_j^2 is the R-squared from such a regression
- $TSS_j = \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2$ is the total sample variation in x_j

- For simple linear regression,

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Standard Error of the Estimators

<u>Estimating σ^2</u>

- σ^2 is unknown and needs to be estimated from the data

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} \hat{u}_i^2}{n-k-1} = \frac{SSR}{n-k-1}$$

- $n-k-1$ is the degree of freedom for OLS with *n* observations and *k* independent variables
- $k+1$ is the number of parameters (including one for the intercept)
- The $(k+1)$ *first order conditions* for OLS lose $(k+1)$ degrees of freedom

<u>Standard Error of $\hat{\beta}$</u>

$$se(\hat{\beta}_j) = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^{n} \hat{r}_{ij}^2}} = \sqrt{\frac{SSR/(n-k-1)}{SST_j(1-R_j^2)}}$$

t-Test

t-Statistic

$$\frac{\hat{\beta}_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$$

- The *t*-statistics $\hat{\beta}_j / se(\hat{\beta}_j)$ follows a *t* distribution with $(n-k-1)$ degrees of freedom
- As the degree of freedom increases, *t* distribution converges to the standard normal distribution

t-Test (Cont'd)

<u>Used to test if the **null hypothesis** $\beta_j = 0$</u>

- If indeed $\beta_j = 0$, the probability of seeing a very large *t* value is very small;
- If we obtain a very ***large*** *t*-statistic value, most likely $\beta_j = 0$ is **NOT true**!
- Find a ***critical value*** (cutoff) of *t* value, *c*, to ***reject*** $\beta_j = 0$ at a certain ***confidence level*** (e.g., 95%, 99%)

$$\Pr(|\boldsymbol{t}_{\hat{\beta}_j}| > c) = 2 \cdot \Pr(\boldsymbol{t}_{\hat{\beta}_j} > c) = 0.5\ (or\ 0.1)$$

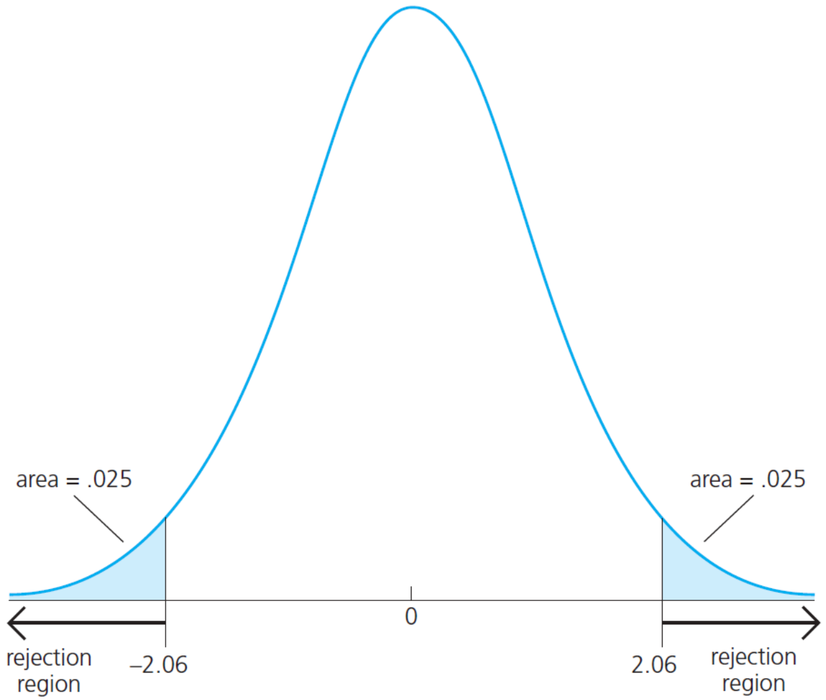
- c is the **$(100 - \alpha/2)$th percentile of t_{n-k-1} distribution** (where α = 100(1-confidence level))

- If the *t*-statistic we obtained is ***larger*** than c in absolute value, $|\boldsymbol{t}| > \boldsymbol{c}$, we can reject the hypothesis $\beta_j = 0$ at the given confidence level (e.g., 95%, 99%)
  - In this case, we say $\hat{\beta}_j$ is *significantly different from zero* (or simply is *significant*);
  - This means x_j has a significant (or nonzero) effect on y.

p-Value

$$p = \Pr(|\boldsymbol{t}_{\hat{\beta}_j}| > t) = 2 \cdot \Pr\left(\boldsymbol{t}_{\hat{\beta}_j} > |t|\right) = 2 \cdot \Pr\left(\boldsymbol{t}_{\hat{\beta}_j} < -|t|\right)$$

- $p < 0.05$ (or 0.01) allows us to reject the hypothesis $\beta_j = 0$ at the confidence level of 95% (or 99%)
  - In this case, we say $\hat{\beta}_j$ is *significantly different from zero* (or simply is *significant*);
  - This means x_j has a significant (or nonzero) effect on y.



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Collinearity

What determines the variance of $\hat{\beta}$?

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{TSS_j(1 - R_j^2)}$$

- R_j^2 is the R-squared from regressing x_j on all the other independent variables
- If x_j can be very well explained by (some of) the other independent variables, R_j^2 will be large, which will lead to a large $Var(\hat{\beta}_j)$

Collinearity (Cont'd)

<u>Multicollinearity</u>

- If one independent variable is highly correlated with one or more other independent variables

- ***Variance Inflation Factor (VIF)*** to assess multicollinearity
  - The ratio of $Var(\hat{\beta}_j)$ when fitting the full model divided by $Var(\hat{\beta}_j)$ if fit on its own
  - VIF exceeding 5 or 10 is a sign of multicollinearity

$$VIF(\hat{\beta}_j) = \frac{1}{1 - R_j^2}$$

Summary

- Variance of the estimators
- *t*-Test
  - *t*-statistic
  - *t*-threshold
  - *p*-value
- Multi-collinearity
  - VIF

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Qualitative Predictors and Dummy Variables

Qualitative Predictors

- Quantitative vs Qualitative Variables
  - Discrete numerical variables vs. Categorical variables
  - Categorical variables: Ordinal vs. Nominal

  - *Discrete numerical* variables: if effects still linear, can be treat as regular numerical variables in the regression
    - E.g., size of family, number of doors of cars
  - *Ordinal categorical* variables: if effects not linear, needs to be treated as qualitative variables with separate levels
    - E.g., elite membership levels: 1,2,3…; education levels
  - *Nominal categorical* variables: must be treated as qualitative variables
    - E.g., color, city, ethnicity

  - In R, *factors* with different *levels* (i.e., possible values)

Predictors with Two Levels

- Qualitative variables with only two levels:
  - Create an indicator or *dummy* variable

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is female} \\ 0 & \text{if } i\text{th person is male} \end{cases}$$

  - Interpret the coefficients correctly

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i\text{th person is female} \\ \beta_0 + \epsilon_i & \text{if } i\text{th person is male} \end{cases}$$

    - β_0: average among males
    - $\beta_0 + \beta_1$: average among females
    - β_1: difference between males and females

Predictors with Two Levels (Cont'd)

- Coding could vary:

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is female} \\ -1 & \text{if } i\text{th person is male} \end{cases}$$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i\text{th person is female} \\ \beta_0 - \beta_1 + \epsilon_i & \text{if } i\text{th person is male} \end{cases}$$

β_0: overall average
β_1: amount females (males) are above (below) average

Predictors with Multiple Levels

- Qualitative variables with more than two levels:
  - Create separate indicator or dummy variables to possible values

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th car is white} \\ 0 & \text{if } i\text{th car is not white} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th car is black} \\ 0 & \text{if } i\text{th car is not black} \end{cases}$$

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i\text{th car is white} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if } i\text{th car is black} \\ \beta_0 + \epsilon_i & \text{if } i\text{th car is red} \end{cases}$$

Predictors with Multiple Levels (Cont'd)

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i\text{th car is white} \\ \beta_0 + \beta_2 + \epsilon_i & \text{if } i\text{th car is black} \\ \beta_0 + \epsilon_i & \text{if } i\text{th car is red} \end{cases}$$

- Interpret the coefficients correctly
  - β_0: average among red cars
  - β_1: difference between white cars and red cars
  - β_2: difference between black cars and red cars

- ***One fewer*** dummy variable than the number of levels
  - Choose one ***baseline*** value/level